

Islamic University of Technology (IUT)

Department of Computer Science and Engineering (CSE)

Assignment 2, Winter Semester 2024-2025

Course Title: Geometry and Differential Calculus

Course No.: MATH 4141, Section No.: SWE

Name:

Student ID.:

1. (a) Remove the first-degree terms from each of the following equations:
 - (i). $x^2 + y^2 + 6x - 8y + 9 = 0$;
 - (ii). $2x^2 + 3y^2 + 4x + 6y - 5 = 0$;
 - (iii). $3x^2 + 2y^2 - 12x + 10y + 7 = 0$.
- (b) For each of the following equations, determine the appropriate rotation formula so that once the rotation of axes is made the equation contains no xy -term. Also, sketch and analyze the new equations.
 - (i). $x^2 + 6xy + y^2 + 4x - 3y + 1 = 0$;
 - (ii). $4x^2 - 4xy + y^2 + 8x + 2y = 0$;
 - (iii). $x^2 - 8xy + 4y^2 - 6x + y + 2 = 0$.
2. (a) Determine whether the second-degree equation $2x^2 - 5xy + 2y^2 + 3x - y - 6 = 0$ represents a pair of straight line or not. If so, then
 - (i). Separate the lines, and find their point of intersection and angle between them.
 - (ii). Find the equation of the bisectors of the angles between the lines.
 - (iii). Plot the lines in the same set of axes using different styles.
 - (iv). Find the area of the triangle formed by the lines and x -axis.

- (b) Find the value of λ for which the general equation of second degree $x^2 + \lambda xy + y^2 + 2x - 4y + 1 = 0$ represents pair of straight lines. Also, find the point of intersection and the angle between them.
- (c) Find the value of k , if the lines joining the origin to the points of intersections of the curve $x^2 + xy + y^2 - 4x + 2y + 3 = 0$ and $x + y = k$ are mutually perpendicular. Verify this conjecture graphically.
3. (a) Find the direction cosines of the line that is drawn from the origin to the point $(3, -2, 6)$.
- (b) Find the distance of the point $P(1, -2, 3)$ from the line through $Q(2, 1, -1)$ whose direction cosines are proportional to $2, -1, 2$.
- (c) Find the projection of the joining of two points $(1, 2, -3)$ and $(4, -1, 2)$ on the line through the points $(0, 1, 2)$ and $(3, 2, -1)$.
4. (a) Find and sketch the equation of the plane that passes through the points $(1, 0, 2)$, $(2, 1, 3)$, and $(0, 2, 1)$.
- (b) Find the distance between the plane found in (a) and the point $(3, -1, 4)$ and explain it geometrically.
- (c) Show that the four points $(1, 2, 3)$, $(2, 3, 4)$, $(3, 5, 7)$, and $(4, 7, 11)$ are coplanar.
- (d) Show that the planes $2x - y + z = 3$ and $x + 2y + 2z = 5$ are perpendicular to each other.
- (e) Find the acute angle between the planes $x + y + z = 1$ and $2x - y + 2z = 3$.
5. (a) Find the equation of the plane passing through the intersection line of the planes $x + y + z - 2 = 0$ & $2x - y + z = 1$ and the point $(1, 1, 1)$.
- (b) Find the equation of the plane which is parallel to the plane $3x - y + 2z - 4 = 0$ and at a distance of 5 units from the point $(2, -1, 3)$.

6. (a) Show that the lines $\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-1}{4}$ and $\frac{x-3}{1} = \frac{y-1}{2} = \frac{z+2}{3}$ are coplanar. Also, find their point of intersection and the equation of the plane containing them.

(b) Find the shortest distance (SD) between the lines $\frac{x-2}{1} = \frac{y+1}{2} = \frac{z-3}{-1}$ and $\frac{x+1}{2} = \frac{y-2}{-1} = \frac{z+1}{3}$. Hence, find the equation of the SD line and the points at which it meets the given lines.

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